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Persuasion through Slanted Language: Evidence from the Media Coverage of Immigration†

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I study the persuasive effects of slanted language, exploiting a ban on the politically charged term "illegal immigrant" by the Associated Press (AP) news wire. My empirical strategy combines the timing of the ban with variation across media outlets in their baseline reliance on AP copy. I document sizable diffusion of the ban from AP copy to media outlets. Moreover, individuals exposed to the ban through local media show significantly lower support for restrictive immigration policies. This effect is more pronounced for moderates and in locations with fewer immigrants, and does not transfer to views on issues other than immigration. (*JEL* D72, L82, Z13)

In the United States, Republicans and Democrats often use strikingly different language when they talk about policy-relevant issues. Typically, Republican politicians and right-leaning media talk about "illegal immigrants," "gun control," and the "death tax." Democratic politicians and left-leaning media refer to the same issues as "undocumented immigrants," "gun safety," and the "estate tax." There is ample anecdotal evidence that strategists from both sides perceive this use of language as instrumental for promoting policy positions favorable to their platforms (Luntz 2007; Lakoff and Ferguson 2006; GOPAC 1990). If language has a causal impact on how people think about these issues, such partisan tactics could contribute to polarized public perceptions of the same factual reality (Alesina, Miano, and Stantcheva 2020).

Yet evidence on the persuasiveness of slanted language i.e., on whether it can indeed sway the views of the public in the intended direction is lacking. From an empirical standpoint, this question is challenging for at least two reasons. First, besides nonpecuniary political incentives, profit-maximizing media outlets have an incentive to slant their language to appeal to the views of their audience. This makes it hard to disentangle cause and effect. Second, slanted language can be accompanied by other politically motivated editorial choices, ranging from selective emphasis

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on certain issues (Puglisi and Snyder 2011) to explicit endorsements (Chiang and Knight 2011). Such choices may independently affect the views of the audience.

To overcome this challenge, I propose a supply-side source of variation in media slant. I take advantage of the fact that US media outlets source some of their content from the Associated Press (AP)—a news wire agency that gathers and distributes news to member outlets. Since the AP distributes a single news feed to all members, its aim is to produce neutral coverage that has broad appeal to outlets of any ideology (Fenby 1986). This philosophy has led to strict and rigid guidelines for the use of politically sensitive language.

I exploit an abrupt reversal in AP guidelines regarding a specific politically charged term, "illegal immigrant." In April 2013, after years of resisting demands to revise its guidelines, the AP switched from officially *recommending* "illegal immigrant" to refer to people living in the United States without legal authorization to *banning* its use in AP dispatches.

The ban happened at a time when the issue of immigration and the language used to talk about it were highly politicized. Figure 1 illustrates the partisan divide in use of the term "illegal immigrant" in political speech and in the media.[1] In Congress, Republican representatives use the label "illegal" about 50 percent of the time they use the word "immigrant," while this frequency is less than 5 percent among Democrats. Similarly, the term appears twice as frequently in the right-leaning *Fox News* and *Washington Times*, compared with the left-leaning *MSNBC* and *Washington Post*.

Beyond the clear ideological charge of the banned term, the setting has several features that make it attractive for studying the causal effects of slanted language. First, the AP is structured as a cooperative owned by its members, and member outlets span the ideological spectrum. It would therefore be hard for demand from the audience of any individual outlet to affect AP's style decisions. If this premise is correct, the ban produces variation in an input to the editorial production function that is orthogonal to outlet-specific demand.

Second, the ban does not appear to be politically motivated. According to the AP, the decision was made for consistency with a recent policy of avoiding labeling people that was implemented in other contexts. For example, about a month prior to banning "illegal immigrant," the AP banned the term "schizophrenic" and replaced it with "diagnosed with schizophrenia." The AP's accounts are supported by the data: I find no pretrends in use of the term "illegal immigrant" or in the immigration slant of AP dispatches.

Third, I find that media outlets differ greatly in the extent to which they rely on AP copy. Even though the ban is an aggregate shock to the entire media landscape, differences in AP reliance across outlets create variation in exposure to this shock. I can therefore implement an empirical strategy of comparing outlets with different baseline reliance on AP copy—i.e., different *AP intensity*—and the views of their respective readers before versus after the ban.

[1] The reason for this divide can be traced to deliberate party strategy. For example, the term "illegal immigrant" was advocated by Republican strategist Frank Luntz. Luntz has urged Republicans to always use the term "illegal immigrant" and to emphasize on border security, calling the linguistic distinction between "illegal immigrant" and "undocumented immigrant" the "political battle of the decade" (Luntz 2007).

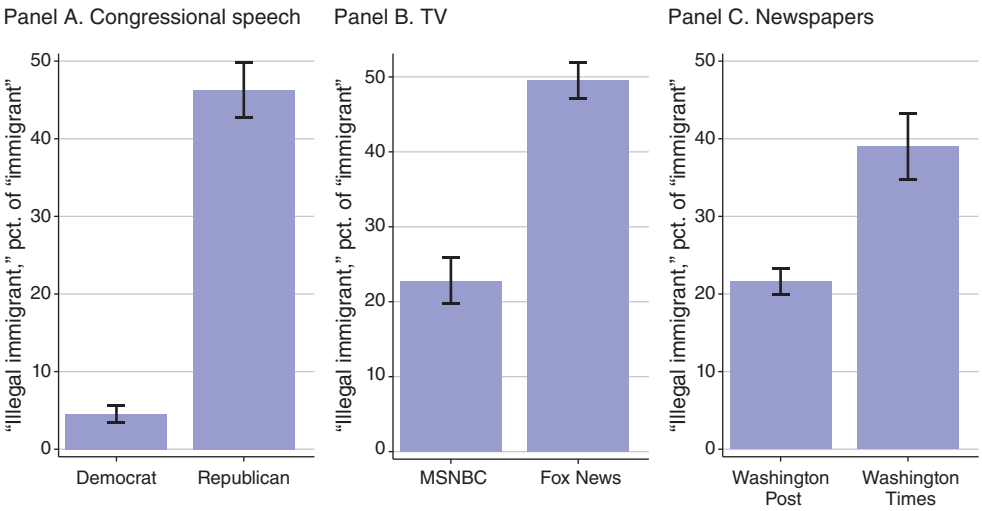

Figure 1. Ideological Charge of the Term "Illegal Immigrant"

Notes: Figure shows use of the term "illegal immigrant" relative to the term "immigrant" in political speech and in the media. Panel A: congressional speech, comparing Democratic and Republican speakers. Panel B: Cable TV, comparing MSNBC and Fox News. Panel C: newspapers, comparing the *Washington Post* and the *Washington Times*. The data cover the years 2009 to 2017.

Sources: Congressional Record (Gentzkow et al. 2018); GDELT TV Archive (Internet Archive 2009-2017); and ProQuest (ProQuest 2009-2017), respectively

I start my analysis by documenting how the ban affected the AP's coverage of immigration, using the text of all immigration-related AP dispatches released between 2009 and 2017. I find that, as intended, the ban caused the term "illegal immigrant" to instantaneously disappear from the AP's feed. At the same time, the volume of immigration coverage and other dimensions of the AP's slant and sentiment remained largely unchanged. As a substitute for the label "illegal," the new guidelines suggested the phrases "living in the county illegally" or "without legal permission." However, text analysis reveals that these reformulations compensated for only a fraction of the decline in use of the banned term. Hence, a significant part of the treatment in this natural experiment consists of substitution from "illegal immigrant" to "immigrant," without any reference to legal status.

In the second part of the paper, I track how the ban diffused from the AP's feed into the language used by media outlets across the United States. In this analysis, I use text data from about 2,400 print and online outlets, including a subsample of 850 daily print newspapers. I employ a generalized difference-in-differences strategy comparing use of the term "illegal immigrant," normalized by use of the term "immigrant," before and after the ban in media outlets with different AP intensity. Specifically, I measure the AP intensity of each outlet as the share of pre-ban articles using the term "immigrant" that either credit the AP explicitly or are flagged by a plagiarism-detection algorithm as highly similar to a recent AP dispatch.

My results suggest a large degree of diffusion: 1-standard-deviation higher AP intensity is associated with a 15 percent decline in use of the term "illegal immigrant" after the ban. Put differently, outlets that source between 13 percent and 85 percent of their articles from the AP (i.e., ones in the top quartile of positive AP

intensity) reduce their use of the term by 50 percent, compared with outlets that do not use the AP at all. This effect is driven mostly by articles sourced from the AP as opposed to original articles.[2] A placebo test that exploits the intensity of another major news wire, Reuters, confirms that these results are not driven by general differences between outlets that rely more or less on news wire copy.

Given the strong political charge of the term "illegal immigrant," we might expect differential reactions by left-leaning outlets, for which the ban is ideologically congruent, compared with right-leaning outlets, for which it is incongruent. Indeed, I find a stronger degree of diffusion for left-leaning outlets compared with right-leaning ones, but use of the term "illegal immigrant" declines significantly for both.

In the third part of the paper, I study the impact of the ban on public views on immigration policy, using pre- and post-ban waves of a large electoral survey: the Cooperative Congressional Election Study (CCES). Here I focus on daily print newspapers and use data on their circulation across US counties. To estimate the reduced-form effect of the ban, I employ a generalized difference-in-differences strategy comparing CCES responses before and after the ban in counties with different AP intensity of locally circulated newspapers. Alternatively, to scale magnitudes in terms of exposure to the term "illegal immigrant," I instrument its frequency by the interaction of county-level AP intensity and the timing of the ban. This strategy accounts for time-invariant county characteristics correlated with AP intensity, but relies on the assumption that their effect on respondents' views did not change at the time of the ban. I address this threat by showing that the estimates are stable to three sets of demanding controls: (i) a range of baseline county characteristics interacted with time, (ii) fixed effects that absorb any state or media-market level shocks, and (iii) county-specific linear trends absorbing any confounds that vary by county but evolve smoothly over time.

The results suggest that 1-standard-deviation higher AP intensity of locally circulated media is associated with 0.8 percentage point lower support for increasing border security after the ban. This corresponds to 2 percent of the gap in support for border security between Republican and Democratic respondents. It implies a persuasion rate—i.e., share of readers exposed to the treatment who changed their position (DellaVigna and Kaplan 2007; DellaVigna and Gentzkow 2010)—in the range of 1.9 percent to 4.4 percent. These estimates are robust to controls, are not driven by pretrends, and do not reflect general reliance on news wire content as measured by Reuters intensity.

I observe a similar shift in support for restricting immigration in four out of the five policy questions I am able to track across pre- and post-ban survey waves, as well in an index that aggregates all immigration-related CCES questions, including rotating ones. Specifically, I find significant effects on support for increasing border security, allowing police to question suspected unauthorized immigrants, fining firms that employ them, prohibiting their access to public services, and no significant effect on support for legalization.

[2] I also find the same gap in reference to legal status as documented for AP dispatches and the same null effects on volume of immigration coverage and on other aspects of slant and sentiment.

Moreover, a falsification exercise that examines views on other politically divisive policies suggests that these effects are specific to immigration policy. I find no effect on the responses to any of the other policy-related questions in the CCES, which range from abortion and gay marriage to healthcare and economic policy.[3]

While the above results apply to the sample of all CCES respondents, they are more pronounced for the 33 percent of respondents who self-report reading a print newspaper frequently (i.e., did so the day before the survey) compared with respondents who did not read a newspaper (p-value $= 0.13$). The effects are also more pronounced for self-described moderates compared with conservatives or liberals (p-value $= 0.22$) and for respondents living in places with a lower immigrant population (p-value $= 0.05$). These patterns of heterogeneity are consistent with a persuasion mechanism: news consumers with moderate ideology and those with less direct exposure to immigrants are likely to have weaker priors on the issue, and hence hold views more susceptible to media framing.

This paper contributes to several streams of literature. First, it relates to research on slant in the language used by media and politicians (Groseclose and Milyo 2005; Gentzkow and Shapiro 2010; Gentzkow, Shapiro and Taddy 2019; Card et al. 2022), which has focused on the measurement of slant and on its determinants. Importantly, Gentzkow and Shapiro (2010) show that consumers' attitudes explain about 20 percent of the variation in the slant of US newspapers. Here I study the reverse direction of causality—from exposure to slanted language toward the political views of readers—and separate this channel from the tendency of media outlets to serve consumers' preference for like-minded content.

The paper also contributes to a literature on the effects of media on political attitudes and outcomes. The majority of this research exploits quasi-random or experimentally manipulated variation in access to a particular media outlet to estimate its causal effects (DellaVigna and Kaplan 2007; Martin and Yurukoglu 2017; Enikolopov, Petrova, and Zhuravskaya 2011; Durante, Pinotti, and Tesei 2019; Gerber, Karlan, and Bergan 2009; Levy2021). By design, the treatment in this strategy consists of the bundle of editorial choices that differentiate the outlet of interest from alternative sources of information. Fewer studies are focused on a specific mechanism of media persuasion e.g., the direct endorsement of candidates for office (Chiang and Knight 2011) or the volume of coverage of a politician or a politically sensitive issue (Snyder and Strömberg 2010; Puglisi and Snyder 2011; Schneider-Strawczynski and Valette 2021). This paper fits into the second category, but differs by studying an arguably more subtle mechanism: slanted language.[4]

My findings suggest that views on immigration policy can be sensitive to how the issue is framed. This lines up with recent evidence of widespread misinformation regarding the characteristics of immigrants and the effects of immigration (Alesina, Miano, and Stantcheva 2022; Haaland and Roth 2020; Grigorieff, Roth,

[3] On net, the relatively small and localized effect on immigration policy views appears (on average) insufficient to shift party support in national elections and I detect no effect on voting or intent to vote for Republican candidates reported in the CCES. I also find no effect on Republican vote share in electoral data.

[4] It is not clear a priori how the possible persuasive effect of slanted language might compare with that of more obvious biases. On the one hand, slanted language arguably represents a much milder treatment. On the other hand, more direct biases may be easier for consumers to notice and discount by either switching away from the biased media outlet (Durante and Knight 2012) or by taking its ideological stance into account when making political choices (Chiang and Knight 2011).

and Ubfal 2020; Couttenier et al. 2021; Keita, Renault, and Valette 2021). Whereas in experimental settings, information treatments that aim to correct misperceptions have shown generally small effects, there is some evidence that narrative or "framing" treatments may be more effective (Bursztyn and Yang 2022; Alesina, Miano, and Stantcheva 2022).

Framing effects, which occur when differences in the presentation of an issue affect individuals' responses, are themselves subject to a large literature in communication, social psychology and economics (Strömberg 2015; Scheufele and Tewksbury 2007; Chong and Druckman 2007; Moreno-Medina et al. 2022).[5] Closest to this paper are survey experiments that randomize respondents' exposure to various immigration-related frames (McCabe, Matos, and Walker 2021; Merolla, Ramakrishnan, and Haynes 2013; Haynes, Merolla, and Ramakrishnan 2016; Ommundsen et al.2014; Knoll, Sanborn, and Redlawsk 2011; Rucker, Murphy, and Quintanilla 2019) and evaluate their effects on opinion or policy views. I use a different methodology based on a natural experiment, and provide large-scale observational evidence of framing effects "in the wild." This allows me to study the effects of persistent exposure to different framing and examine the heterogeneity of these effects.

Finally, the large magnitude of diffusion of the AP's ban documented in the first part of the paper implies significant power of news wire agencies to influence media slant. This complements evidence in the context of French online media (Cage, Herve, and Viaud 2020).

The rest of the paper is organized as follows. Section I, I discuss the details of the ban and analyze how it affected the AP's coverage of immigration. In Section II, I track the diffusion of the ban into the language used by media outlets across the United States. In Section III, I analyze the effect of the ban on public views on immigration policy.

I. The Ban and Its Effect on AP's Language

A. *Background*

Motivations for the Ban.—The term "illegal immigrant" was dropped from the AP guidelines on April 3, 2013 (Abad-Santos 2013). The decision was surprising because the AP had previously resisted pressures to change its language on the issue. For example, back in 2010 the AP reaffirmed "illegal immigrant" as the preferred term to refer to unauthorized immigrants, rejecting the demands of the advocacy group "Define American" (Poynter 2011). The alternative endorsed by the left ("undocumented immigrant") was not sanctioned because the AP considers it legally inaccurate; this continues to be the case to this day.

As a reason for the sudden reversal, the AP named a recent policy of not labeling people, which was being implemented in other areas (e.g., replacing the term

[5] A related literature studies the relationship between linguistic features and economic behaviors across societies (Ginsburgh and Weber 2020; Chen 2013). More broadly, a long-standing hypothesis called "linguistic relativity" states that linguistic categories influence cognitive categories and perceptions (Winawer et al.2007).

Table 1—AP's Ban on "Illegal Immigrant"

Panel A. AP Stylebook guidelines before and after the ban	
Pre-Ban	*Post-Ban*
illegal immigrant Used to describe those who have entered the country illegally, it is the preferred term, rather than illegal alien or undocumented worker. Do not use the shortened term illegals.	**illegal immigration** Entering in a country in violation of civil or criminal law. Except in direct quotes essential to the story, use illegal only to refer to an action, not a person: illegal immigration, but not illegal immigrant. Acceptable variations include living in or entering a county illegally or without legal permission. Expect in direct quotations, do not use the terms illegal alien, an illegal, illegals or undocumented. Do not describe people as violating immigration laws without attribution. […]
Panel B. Example of AP dispatches before and after the ban	
March 18, 2013: two weeks pre-ban	April 16, 2013: two weeks post-ban
Senate Panel OKs letting non-citizens, including illegal immigrants, get driver's licenses ST. PAUL, Minn. (AP)—Bills that would let illegal immigrants get a Minnesota driver's license are moving forward at the Capitol. The Senate Transportation and Public Safety Committee on Monday passed a bill to ease restrictions on driver's licenses for non-US citizens. A House committee endorsed a similar bill last week. Sen. Bobby Joe Champion, a Minneapolis Democrat, says his bill would make Minnesota roads safer by funneling more drivers through the state's driving test and making it easier for them to buy automobile insurance. Republicans say the change could lead to unintended consequences, like illegal immigrants using state IDs to vote. The bill passed 10–7, with all Democrats in favor and all Republicans voting against it.	**Immigrant driver's license bill takes step forward in Oregon Senate committee work** SALEM, Ore. (AP) – An Oregon Senate committee has advanced a bill granting four-year driver's licenses to people who can't prove they're legally in the United States. The Senate Business and Transportation Committee approved the measure Monday on a 4-2 vote. The bill would allow immigrants who have lived in Oregon for at least a year and meet other requirements to apply for driver's cards without proving legal presence. The card would be valid for only four years– half as long as a standard Oregon license– and would state "driving privilege only." Supporters say it will make Oregon roads safer because there would be fewer untrained and uninsured drivers, but opponents say it could create a culture of crime in the state. The bill goes to a legislative budget committee.

Notes: Panel A: Extracts from the *AP Stylebook* entry on illegal immigration before and after the ban. Panel B Extracts: examples of AP dispatches before and after the ban (Associated Press 2010, 2014).

"schizophrenic" with "diagnosed with schizophrenia").[6] According to a statement released by AP executives, the discussion about labeling people, instead of behavior, and the need for consistency with the new policy led them to reconsider the term "illegal immigrant."[7]

Table 1, panel A presents the exact wording of the AP guidelines before and after April 2013. Since "illegal immigrant" was now banned, the new guidelines suggested the following substitutes: "living/entering the county illegally/without legal permission." Yet in their statement, AP executives recognized the possibility that those alternatives might "make it harder for writers" compared with the concise label "illegal." The ban took effect immediately in the AP's online guidelines, which are also embedded in text editors.

[6] Other examples of the AP's de-labeling policy, also referred to as "person-first" language, concern labels related to disabilities ("disabled," "handicapped"); age ("the elderly"); addiction ("addict," "alcoholic"); and homelessness ("the homeless"). Since these changes are not related to immigration, I assume that they do not affect views on immigration policy.

[7] The AP's full statement on the ban is available at https://blog.ap.org/announcements/illegal-immigrant-no-more.

The Associated Press.—The AP is a not-for-profit cooperative of 1,300 US newspapers and broadcasters owned by the outlets it serves. AP member outlets elect its board of directors annually for three-year terms, with a term limit of nine years.[8] This organizational structure suggests that AP decisions balance the interests of all member outlets.

Member outlets have access to AP-produced news and photos, share costs, and contribute content. "Regular" membership, which applies to all print newspapers published in the United States, also entails permission for the AP to distribute local news reports produced by the member outlet (Associated Press 2016).[9] The AP's feed thus consists of two components: (i) original work produced by AP journalists and (ii) rewrites of stories produced by member outlets.

While the AP is the main news wire used in US media, Reuters—an international for-profit news agency based in the United Kingdom and owned by the Thomson Reuters Corporation—offers competing services and is also used by some US newspapers (Poynter 2014). Reuters did not implement any changes in its rules regarding the term "illegal immigrant." Therefore, notwithstanding differences between the two news agencies, in some parts of the analysis I use Reuters as the closest available placebo for AP.

B. *Data*

To shed light on the nature of the treatment in this natural experiment, I collect the text of all dispatches released by the AP between June 2009 and June 2017 that use the term "immigrant." Specifically, I search the *Factiva* database (Factiva 2009–2017) for the keyword "immigrant" (singular or plural), restricting the source to Associated Press Newswires.[10] This results in 28,000 dispatches, 8,000 of which use the term "illegal immigrant." I record the date, headline, and full text of each dispatch.

C. *Example and Descriptive Evidence*

I start with an example of how the ban affected the AP's language on immigration, even when it comes to covering the same issue. Table 1, panel B presents two reports on state laws on immigrants' drivers licenses: one in Minnesota, released

[8] The board has about 20 members, most of whom represent media groups consisting of dozens of outlets. In 2012, the members of the board had the following affiliations: Rust Communications, Lakewat Publishers, Cowles Company, E.W. Scripps Company, Lee Enterprises, Advance.net, Community Newspaper Holdings, The New York Times Company, Ogden Newspapers, *The Santa Fe New Mexican* and the *Taos News, The Dallas Morning News*, Schurz Communications, Times Publishing Company, *Milwaukee Journal Sentinel*, Hearst Corporation, Washington Post Media, Paxton Media Group, Belo Corp.

[9] Most US newspapers are, by default, members of the AP. The *Editor and Publisher* yearbook lists the AP under "news wire services" for about 96 percent of the daily print newspapers included in my sample. This lack of variation prevents me from using membership as a proxy for exposure to the ban. Yet, as shown in Section II, I find large variation at the intensive margin of actual use of AP copy.

[10] In defining the corpus, I focus only on dispatches that use the term "immigrant" and do not consider other immigration-related keywords (such as "immigration"). This is because "illegal immigrant" is the only term banned by the AP. In contrast, "illegal immigration" was explicitly exempt from the ban because it labels an action and not a person (Table 1, panel A). On the other hand, my definition may be too broad if it captures reporting on non-US immigration. I find that this is rare: meta-data from *Factiva* suggests that only 8 percent of the dispatches using the term "immigrant" refer to non-US immigration.

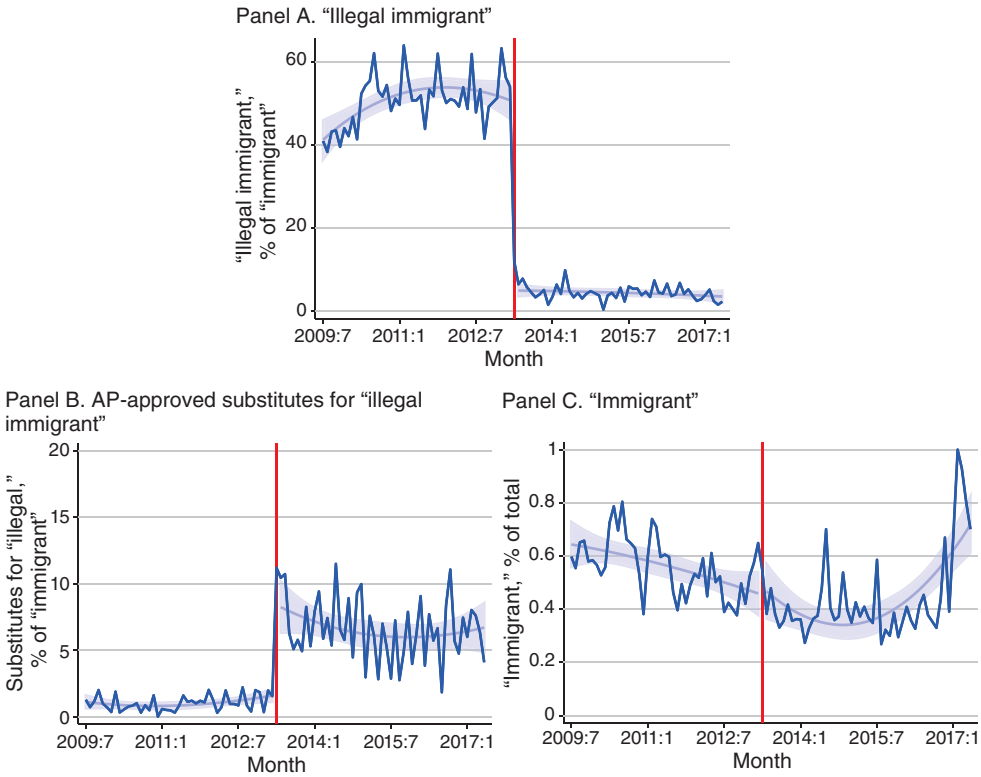

Figure 2. The AP's Language on Immigration over Time

Notes: The figure shows monthly time series. Panel A: Use of the term "illegal immigrant" relative to the term "immigrant." Panel B: Use of the AP-approved substitutes "enter* / live* in the country illegally / without legal permission" relative to the term "immigrant." Panel C: Use of the term "immigrant" relative to total dispatches.

about two weeks before the ban, and one in Oregon, released two weeks after. Both dispatches have a neutral tone and state arguments for and against the law. However, the headline of the first dispatch refers to licenses to "illegal immigrants," while the second headline refers to an "immigrant drivers' license bill," arguably changing the framing of the story.[11]

The timing of this change in language coincides precisely with the announcement of the ban. Figure 2, panel A shows the monthly number of AP dispatches that use the term "illegal immigrant" relative to the number of dispatches that use the term "immigrant." This share drops from an average of 50 percent in the period before April 2013 to less than 5 percent after, suggesting close to perfect compliance with the ban.[12] Panel B of Figure 2 shows that use of the substitutes suggested by the AP guidelines increases sharply after the ban, but the magnitude of this increase is

[11] Online Appendix Figure A3 shows that this is not an isolated case by analyzing the headlines of all dispatches that use the term "immigrant." The label "illegal" clearly features prominently before the ban and virtually disappears after.

[12] Online Appendix Figure A5 shows a similar, though attenuated decline in use of the label "illegal" when the term "immigration" is considered in addition to "immigrant." This is consistent with the fact that, in contrast to "illegal immigrant," the term "illegal immigration" was not affected by the ban (Table 1, panel A).

relatively small (6 percentage points) and insufficient to compensate for the large decline in use of the term "illegal immigrant" (47 percentage points). At the same time, panel C suggests no change in the volume of dispatches using the term "immigrant." Online Appendix Figure A4 repeats this analysis, focusing on headlines, and shows virtually no use of the term "illegal immigrant" after the ban, and virtually no use of the substitutes proposed by the AP.

Online Appendix Figure A6 compares use of the term "illegal immigrant" by the AP with that by Reuters. For Reuters, the share of dispatches using the term "illegal immigrant" relative to the term "immigrant" ranges between 20 percent and 40 percent. While there is a decline in this share over time, its magnitude is small relative to the decline for the AP.

D. *Text Analysis Results*

Changes in the AP's Language.—I adopt two approaches to capture in a comprehensive way any other changes in the AP's language on immigration.

The first approach is to examine how often each of the unigrams from the AP corpus occurs in the same article as the term "immigrant." Figure 3, panel A plots the results, with pre-ban rates of co-occurrence on the x-axis and post-ban rates of co-occurrence on the y-axis. "Illegal" is a clear outlier from the 45-degree line; its association with "immigrant" drops from 0.66 before the ban—the highest among all unigrams in the corpus—to 0.21 after the ban. Yet the figure also suggests that no other unigram compensates for this decline. The closest candidate is "illegally": Its association with "immigrant" increases significantly, but by less than one-half of the decline in "illegal."

The second approach is to ask which words and phrases are predictive for whether an AP dispatch was published before or after the ban. Let $f_{pl,before}$ and $f_{pl,after}$ denote the total number of times phrase p of length l (one to 3 words) is used before and after the ban, respectively. Let $f_{\sim pl,before}$ and $f_{\sim pl,after}$ denote the total occurrences of length-l phrases that are not phrase p—before and after the ban, respectively. Pearson's χ^2 statistic for phrase p of length l is then

$$(1)\quad \chi^2_{pl} = \frac{\left(f_{pl,before} f_{\sim pl,after}\, f_{\sim pl,after} f_{\sim pl,before}\right)^2}{\left(f_{pl,before} + f_{pl,after}\right)\left(f_{pl,before} + f_{\sim pl,before}\right)\left(f_{pl,after} + f_{\sim pl,after}\right)\left(f_{\sim pl,before} + f_{\sim pl,after}\right)}.$$

Figure 3, panel B presents the words and phrases with highest χ^2. "Illegal," "illegal immigrant," and "illegal immigrants" clearly emerge as most diagnostic of publication date. "Illegally" on the other hand has only one-fourth the predictive power of "illegal." The figure also suggests no other substitute for "illegal immigrant" emerging after the ban. Hence, both exercises confirm that synonyms that signal legal status were only partially adopted, and show that no other single phrase replaced the banned term.

The AP's Slant on Immigration.—Was the ban on "illegal immigrant" accompanied by a more liberal slant in the AP's immigration coverage? To examine this

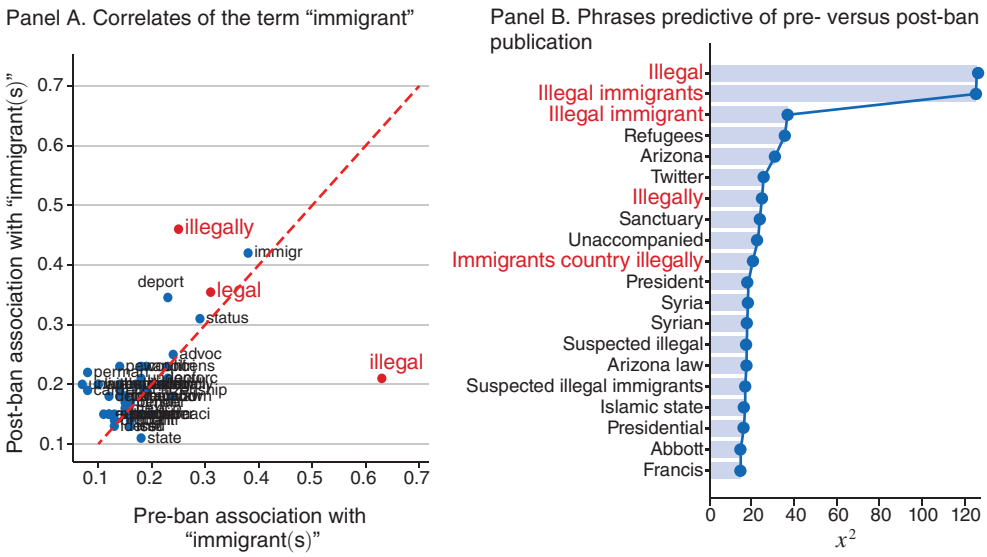

FIGURE 3. CHANGES IN THE AP'S LANGUAGE BEFORE VERSUS AFTER THE BAN

Notes: Changes in the words and phrases used by the AP in immigration coverage before versus after the ban. Panel A: Top 50 unigrams with highest frequency in the text of dispatches using the term "immigrant." Pre-ban frequency on the x-axis; post-ban frequency on the y-axis. Panel B: Phrases most predictive for whether an AP dispatch using the term "immigrant" is published before or after the ban. Top 20 n-grams (n ∈ 1,2,3) based on χ^2 test statistic.

question, I compute a measure of immigration-specific slant based on the similarity of the AP's language to that used by Republicans versus Democrats in Congress. The procedure follows Gentzkow and Shapiro (2010) by first selecting a set of phrases most predictive of partisanship in Congress and then measuring their relative occurrence in AP text.[13] I compute two versions of this index that include and exclude the term "illegal immigrant" and its substitutes. Online Appendix A.1 describes the method in more detail and validates the resulting index.

Figure 4, panel A shows the evolution of the slant index over time. The version that includes the term "illegal immigrant" closely follows the trend in the AP's use of the term. However, once "illegal immigrant" and its substitutes are excluded and I focus on other dimensions of language, I find a flat trend and no change in slant.

The AP's Sentiment on Immigration.—Was the ban on "illegal immigrant" accompanied by a more positive tone in the AP's immigration coverage? To examine this possibility, instead of classifying Republican and Democratic phrases, I use a rule-based method that builds on a lexicon of annotated adjectives (Loria 2018) (see online Appendix A.2). The evolution of the resulting sentiment polarity index, presented in Figure 4, panel B, mirrors that of slant. Sentiment shifts in a positive direction at the time of the ban, but this change is almost entirely accounted for by the absence of the negatively charged adjective "illegal."

[13] Online Appendix Table A1 presents the top partisan phrases that enter the slant measure. The top Republican phrases include "illegal immigrant," "secure border," and "drug cartels." The top Democrat phrases include "undocumented immigrant," "domestic violence," and "hate crime."

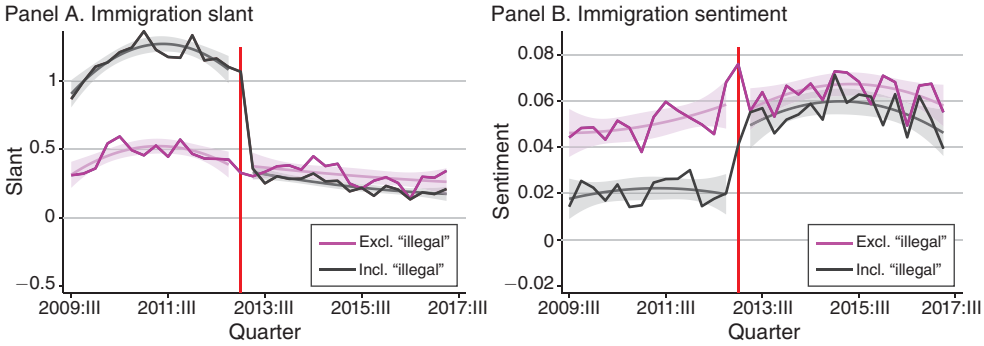

FIGURE 4. THE AP'S SLANT AND SENTIMENT ON IMMIGRATION OVER TIME

Notes: The AP's slant and sentiment on immigration over time, computed including or excluding the term "illegal immigrant" and its substitutes. The figure shows time series by quarter. Panel A: Slant index, measuring the similarity of the language of AP dispatches to the language used by Republicans versus Democrats in Congressional speech. The index ranges from –3 (most left-leaning outlet in the sample) to +3 (most right-leaning outlet in the sample). Panel B: Sentiment polarity index, measuring positive versus negative sentiment in AP dispatches. The index ranges from –1 (negative) to +1 (positive).

Changes by Topic.—To rule out the possibility that these results reflect an over-time shift in immigration-related topics, I implement a Latent Dirichlet Allocation (LDA) topic model (Blei, Ng, and Jordan 2003). I use the model to classify AP dispatches into five topics that can be labeled as follows: law enforcement, legislation, immigrants' integration and social issues, international issues such as the refugee crisis in Europe, and elections. Online Appendix shows that the resulting topic distribution is relatively balanced before and after the ban. Furthermore, most results of the analysis discussed thus far are confirmed *within* topic.

Taking these results together, the analysis of AP text suggests that (i) as intended, the label "illegal" virtually disappears from the AP's feed after the ban; (ii) this is only partially compensated by the substitutes proposed by the AP such that for the most part the ban leads to omission of direct references to legal status; and (iii) other measurable dimensions of the AP's language, volume of immigration coverage, slant, or sentiment did not change with the ban.

II. Diffusion

I next turn to analyzing the diffusion of the AP's ban from AP dispatches into the language of media outlets with different baseline reliance on AP copy. Here I examine two samples: (i) All media outlets for which I can collect content data, including online media and weeklies, and (ii) the subsample of daily print newspapers.

A. *Data*

Media Content: Use of "Immigrant" and "Illegal Immigrant."—My main data source for media content is Newslibrary (NewsLibrary 2009-2017). I focus on print and online outlets that are covered in the database continuously between July 2009

and July 2017; this yields 2566 outlets. For major US newspapers that are missing in NewsLibrary, I use data from ProQuest (ProQuest 2009-2017). This adds 125 newspapers.[14]

I search the databases for the keywords "immigrant" and "illegal immigrant" (singular or plural). The search results in about 1 million articles for "immigrant" and 200,000 articles for "illegal immigrant." I record each article's date of publication, name of the publishing outlet, byline, headline, and text of the first paragraph.[15] Using this information, I compute for each outlet the monthly number of articles using the term "illegal immigrant," normalized by the number of articles using the term "immigrant." This normalization has two goals. First, it accounts for the fact the coverage of NewsLibrary and ProQuest data is not universal, which makes the absolute number of articles somewhat harder to interpret. Second, it allows me to focus on changes in the *language* used to talk about the issue, conditional on the *volume* of coverage devoted to it.

Identifying Articles Sourced from the AP.—I classify an article as sourced from AP if one of two conditions is true: (i) "AP" or "Associated Press" is explicitly included in the byline or the first paragraph (e.g., "according to the AP"), or (ii) a large portion of the text of the article is verbatim identical to the text of a recent AP dispatch. This two-step procedure aims to capture use of the AP as broadly as possible, including cases in which AP copy has been edited or rewritten.

The first criterion looks for cases in which the AP is credited explicitly at the beginning of the article. The same method is employed by Gentzkow and Shapiro (2010) to identify and, in their case, *exclude* news-wire content. Their audit of excluded articles suggests that "virtually all" are indeed wire copy. However, if media outlets use AP copy without explicit credit, this method alone is likely to produce false negatives. Evidence on copying from the French news wire AFP suggests that this may indeed be a common occurrence (Cage, Herve, and Viaud 2020).

Therefore, the second criterion looks at text similarity to recent AP dispatches, regardless of credit. Specifically, I employ a plagiarism-detection algorithm comparing each article to the set of dispatches released by the AP the day before the article is published. I label an article as sourced from the AP if I detect verbatim text overlap $>$ 50 percent. Online Appendix A.3 describes this procedure in detail and discusses summary statistics on AP credit and plagiarism.

AP Intensity.—To proxy media outlets' exposure to the ban, I measure their use of AP copy over the 12 months prior to announcement of the ban. Focusing on the pre-ban period alleviates concerns about changes in the use of AP copy in response to the ban.[16] To rule out anticipation effects, in robustness checks I also consider

[14] The full list of media outlets contained in the sample, along with their corresponding AP intensity and ideology index estimated by Gentzkow and Shapiro (2010), is available at shorturl.at/dtuvM. Note that outlets that never use the term "immigrant" drop out of the sample.

[15] NewsLibrary allows keyword searches showing the first paragraph of all matching articles, but access to the full text is restricted.

[16] Indeed, online Appendix Figure B1 suggests a decline in AP use coinciding with the ban. However, I do not find a particular pattern in this decline by outlet ideology. It is therefore not clear whether this decline is driven by outlets' ideological disagreement with the ban or by other factors. For example, at the start of 2013 Tribune

time windows of 24 to 12 months or 24 to 36 months prior to the announcement of the ban.

I then define *AP intensity* as the share of articles using the term "immigrant" that are sourced from the AP. Online Appendix Figure B2 presents the resulting distribution of AP intensity. The share of AP-sourced articles ranges from zero to 84.6 percent. It is larger than zero for 21 percent of outlets in the full sample of media outlets and for 45 percent of outlets in the sample of print newspapers (online Appendix Table B1). Given this skewed distribution, in the following analysis I use the inverse hyperbolic sine transformation of AP intensity.

Additional Data on Print Newspapers.—For print newspapers, I collect data on the location of headquarters and on baseline circulation from the 2010 *Editor and Publisher* yearbook (Editor & Publisher 2010). This covers the universe of daily print newspapers. I collect data on circulation over time from the historical reports of the the Alliance for Audited Media (2011-2017), which is available for the period 2011–2017 for 380 newspapers in my sample. I also use the index of ideological slant constructed by Gentzkow and Shapiro (2009, 2010), which covers about 370 newspapers in my sample.

B. *Empirical Strategy*

To estimate the rate of diffusion of the ban to the language used by media outlets, I implement a difference-in-differences strategy with continuous treatment. Specifically, I exploit the timing of the ban combined with variation in AP intensity across media outlets. I estimate regressions of the following form:

$$(2) \qquad Illimm/Imm_{mt} = \alpha_m + \beta_t + \rho IHS(APintensity_m) \times PostBan_t + \epsilon_{mt},$$

where $Illimm/Imm_{mt}$ denotes the number of articles using the term "illegal immigrant," relative to ones using the term "immigrant," in media outlet m and month t; IHS$(APintensity_m)$ is IHS-transformed AP intensity measured in the 12 months prior to the ban; $PostBan_t$ is a dummy for post-April 2013; and α_m and β_t are outlet and year-month fixed effects. Standard errors are clustered at the outlet level. To account for the fact that $Illimm/Imm_{mt}$ is imprecisely estimated when the denominator i.e., the number of articles using the term "immigrant" is low—which is a frequent occurrence at the monthly level—I weigh observations by the number of articles using the term "immigrant."

The identifying assumption in this strategy is that use of the term "illegal immigrant" in outlets with different AP intensity would have followed parallel trends in the absence of the ban. To examine the plausibility of this assumption as well as the timing of the effects, I also estimate a dynamic version of the above equation, splitting the dummy for post-ban into a set of half-yearly leads and lags.

newspapers drop the AP for financial reasons (Poynter 2014). In either case, a decline in AP intensity in response to the ban would lead me to underestimate its effects in a difference-in-differences setting.

Table 2—Diffusion of the Ban

	Full sample–print and online			Daily print newspapers			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	"Illegal immigrant," percent of "Immigrant"						
PostBan × IHS(AP-intensity)	−1.613	−1.498	−1.806	−1.365	−1.333	−1.344	−1.756
	(0.171)	(0.160)	(0.217)	(0.207)	(0.186)	(0.219)	(0.265)
Outlet FEs × Linear time trend	No	No	Yes	No	No	No	Yes
DMA × year-month FEs	No	No	No	No	No	Yes	No
State × year-month FEs	No	Yes	No	No	Yes	No	No
Outlet fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year-month fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	139,523	139,506	139,523	66,846	66,784	59,811	66,846
Number of outlets	2,385	2,385	2,385	853	853	785	853
R^2	0.43	0.50	0.49	0.44	0.53	0.62	0.49
Mean dep. var.	20.28	20.28	20.28	21.70	21.66	21.45	21.70
Effect for SD Δ in treatment	−3.53	−3.28	−3.95	−3.35	−3.27	−3.31	−4.31

Notes: Effect of the AP's ban on media outlets' use of the term "illegal immigrant." The level of observation is media outlet × year-month. OLS regressions weighted by the number of articles using the term "immigrant" in all columns. Full sample of print and online outlets in the left panel and subsample of daily print newspapers in the right panel. Standard errors clustered by media outlet.

C. *Results*

Preliminary Evidence.—Before proceeding to estimation of the regression specified in equation (2), I examine use of the term "illegal immigrant" in media outlets with positive versus zero AP intensity. Online Appendix Figure B3 presents the two series. While non AP-intensive outlets appear to gradually decrease their use of the term prior to the ban, for AP-intensive outlets it remains flat and relatively high up until it falls sharply at the time of the ban. This pattern is in line with anecdotal evidence: While the term was becoming more controversial in the years leading up to the ban, it continued to be recommended as "best use" in the AP guidelines.

The figure also suggests that the ban was an aggregate shock: Non AP-intensive outlets also exhibit a decline at the time of the ban, albeit of a smaller magnitude. This may be due to non AP-intensive outlets interpreting the ban as a signal that the term "illegal immigrant" is no longer politically correct. It may also reflect a broader influence of AP copy beyond that captured in my measure of AP intensity.

Both of these features of the data would lead me to underestimate the impact of the ban in a difference-in-differences setting, so that the estimates I provide should be interpreted as a lower bound.

Diffusion Estimates.—Table 2 presents the main regression results corresponding to specification 2, estimated for the full sample of print and online outlets in the left panel, and for the subset of daily print newspapers in the right panel. In both samples, I find a significant reduction in use of the term "illegal immigrant" in AP-intensive outlets relative to less AP-intensive outlets after the ban. The magnitude suggests that 1-standard-deviation higher AP intensity (=2.2 IHS-points) is associated with a reduction use of the term of 3.5 p.p, or about 17 percent relative to the mean (15 percent in the sample of print newspapers).

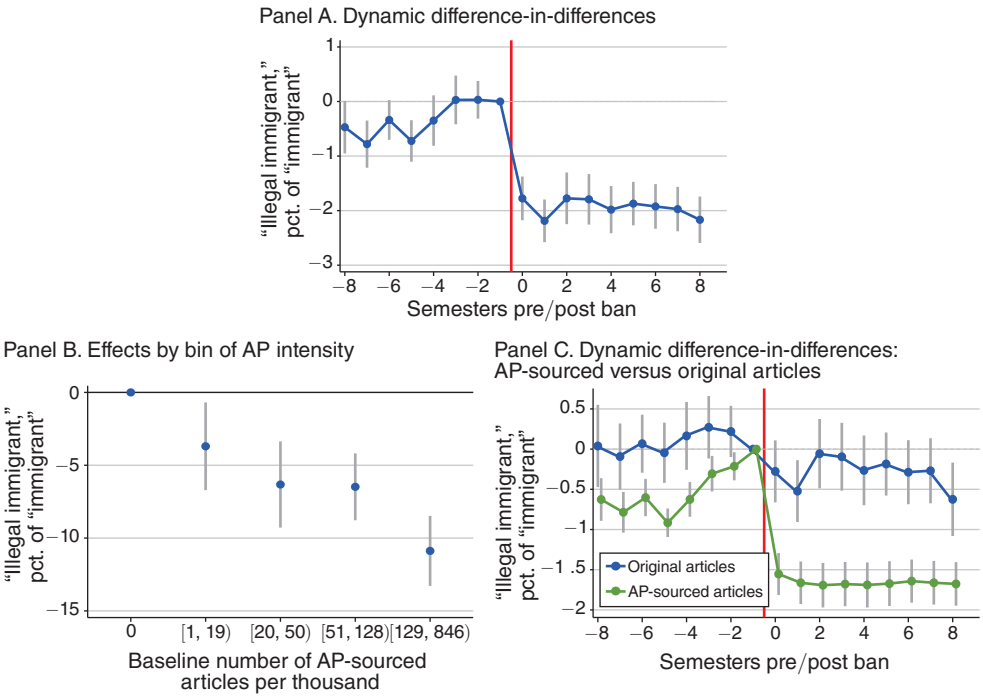

FIGURE 5. DIFFUSION OF THE BAN

Notes: Effect of the AP's ban on media outlets' use of the term "illegal immigrant." Panel A: Regression of media outlets' monthly use of the term "illegal immigrant," relative to the term "immigrant," on the interaction of IHS(AP-intensity) with a set of indicators for semester pre-/ post- ban. The omitted category is the semester before the ban. Panel B: Regression of outlets' monthly use of the term "illegal immigrant," relative to the term "immigrant," on a set of indicators for quartile of (positive) AP intensity interacted with Post Ban. The omitted category is AP intensity = 0. Panel C replicates panel A, splitting the dependent variable into the number of articles that use the term "illegal immigrant" and are sourced from the AP (green) versus ones not sourced from the AP (blue), both relative to the number of articles using the term "immigrant." All regressions control for outlet and year-month fixed effects and are weighted by the number of articles using the term "immigrant." The figures show point estimates and 95 percent confidence intervals. Standard errors are clustered by media outlet. Online Appendix Figure B5 replicates these results for the subsample of daily print newspapers.

The estimates are robust to several controls. Columns 2 and 5 control for year-month × state fixed effects, which absorb the effects of any policy changes or newsworthy events that vary by state. For print newspapers, for which I have information on the exact location of the headquarters, I can further control for finer fixed effects at the level of year-month × designated market area (DMA) (column 6). This restricts the comparison within areas with relatively homogeneous TV and online news consumption profiles and absorbs any shocks that occur at this finer level. Finally, columns 2 and 7 present specifications that control for outlet-specific linear trends. This relaxes the assumption that outlets with different AP intensity follow parallel trends, and identifies the effect of the ban off of the sharp change in use of the term "illegal immigrant" at the time of the ban. The estimates are, if anything, larger in magnitude compared with those from the baseline specification.

To examine the timing of this effect, in Figure 5, panel A I split the interaction of AP intensity and Post-Ban into a set of interactions with half-yearly leads and lags, with the six months before the ban as the omitted category. I find if anything an increasing pretrend, consistent with online Appendix Figure B3. The effect on use

of the term "illegal immigrant" occurs precisely when the ban is announced and is persistent over time, in line with the permanently low supply of AP dispatches using the term "illegal immigrant." Figure 5, panel B suggests a very similar picture for the subsample of daily print newspapers.

In Figure 5, panel B I estimate a flexible specification that interacts each quartile of the distribution of positive AP intensity with Post-Ban, leaving outlets with zero AP intensity as the reference category. The results suggest that the effect of the ban is monotonically increasing in AP intensity. Relative to outlets with zero AP intensity, those in the top quartile—i.e., ones that source between 13 percent and 85 percent of their articles from the AP reduce their use of the term by 10 p.p., or by about percent% relative to the mean.

AP-Sourced versus Original Articles.—If the documented decline in use of the term "illegal immigrant" is linked directly to the change in AP's language, we would expect that it is driven by articles sourced from the AP rather than by original content. To test this, in Figure 5, panel C I split the dependent variable into articles using the term "illegal immigrant" that are sourced from the AP i.e., either credit the AP explicitly or are flagged by plagiarism detection—and ones that are not sourced from the AP. Online Appendix Table B2 presents the corresponding regression results. I find that the diffusion effects are driven mostly by AP-sourced content and the impact on original articles is instead small and, for print newspapers, only marginally significant.

Robustness and Reuters Placebo.—In Table 3, I probe the robustness of the estimates to different definitions of the AP intensity variable. Instead of accounting for both crediting AP and plagiarism, I consider only the share of articles credited to the AP or, alternatively, only the share of articles flagged by plagiarism detection. I also consider a measure based on AP credit in any article and not restricted to articles using the term "immigrant." I find very similar results. In online Appendix Table B3, instead of fixing the pre-ban period to define AP intensity to the 1–12 months before the ban, I shift the measurement further back in time, to 12–24 months or 24–36 months before the ban. The results are robust to this change, which speaks against any selection in use of the AP in anticipation of the ban. The baseline results are also robust to a number of alternative transformations of AP intensity, including binary, log, or IHS with different scaling (online Appendix Table B4).

I next consider a placebo exercise in which I replace AP intensity with Reuters intensity.[17] In contrast to the AP, Reuters did not change its style rules on "illegal immigrant," and while its use of the term declines around 2014 (online Appendix Figure A6), this decline is smaller than that for the AP, and similar to the decline for non-AP intensive media (online Appendix Figure B3). Columns 4 and 8 of Table 3 show no effect when using Reuters intensity as a placebo treatment.

Finally, the estimated decline in use of the term "illegal immigrant" is robust to a number of alternative definitions of the dependent variable—e.g., restricting attention to headlines only, expressing the number of articles using the term "illegal

[17] Since the use of Reuters copy is much more rare in my sample, I express Reuters intensity as the share of Reuters-credited articles per 10,000.

Table 3—Diffusion of the Ban: Robustness and Reuters Placebo

	Full sample–print and online				Daily print newspapers			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	"Illegal immigrant," percent of "Immigrant"							
PostBan × IHS(AP-intensity)	−1.603	−1.562	−1.679		−1.168	−1.433	−1.143	
	(0.187)	(0.185)	(0.289)		(0.218)	(0.211)	(0.334)	
PostBan × IHS(Reuters-intensity)				0.292				0.482
				(0.311)				(0.416)
Method	Credit	Plagiarism	Credit	Credit	Credit	Plagiarism	Credit	Credit
Sample	Imm.	Imm.	All	All	Imm.	Imm.	All	All
Observations	139,523	139,523	138,624	138,624	66,846	66,846	66,362	66,362
Number of outlets	2,385	2,385	2,532	2,532	853	853	852	852
R^2	0.43	0.43	0.41	0.41	0.43	0.44	0.43	0.43
Mean dep. var.	20.28	20.28	20.52	20.52	21.70	21.70	21.85	21.85

Notes: Effect of the AP's ban on media outlets' use of the term "illegal immigrant": Robustness to alternative measures of AP intensity and Reuters placebo. *Method* and *Sample* describe the construction of AP intensity (Reuters intensity) in the respective column. The level of observation is media outlet × year-month. All specification control for media outlet and year-month fixed effects. OLS regressions weighted by the number of articles using the term "immigrant" in all columns. Full sample of print and online outlets in the left panel and subsample of daily print newspapers in the right panel. Standard errors clustered by media outlet.

immigrant" in IHS-transformed levels, using as denominator total articles instead of articles using the term "immigrant," or considering the word "immigration" in addition to "immigrant" (online Appendix Table B5).

Diffusion by Outlet Ideology.—Since media outlets can decide whether and how to use AP content in their editorial production and the banned term was clearly ideologically charged, a natural question is how diffusion of the ban differs by outlet ideology.

To answer this question, I analyze a subsample of about 370 newspapers that I can match to the index of ideological leaning constructed by Gentzkow and Shapiro (2010).[18] I then I estimate the difference-in-differences regression from equation (2) separately for each quartile of the ideology index, standardizing the dependent variable to mean equal to 0 and to standard deviation equal to 1 within each sample.

The resulting difference-in-differences coefficients, presented in Figure 6, panel A, suggest that the ban causes a significant decline in use of the term "illegal immigrant" in 3 out of the 4 samples, but the relative magnitude of the effect differs. The degree of diffusion is almost twice larger for the most left-leaning outlets compared with the most right-leaning ones.

Following the method from Section IIC, I can further decompose this effect into articles sourced from the AP versus presumably original articles. I find remarkably similar declines in use of the term "illegal immigrant" in AP-sourced articles, regardless of outlet ideology (panel B). Instead, the effects on original articles go in opposing directions—a negative effect for left-leaning outlets and a positive effect for center- and right-leaning ones (panel C). This is consistent with a possible spillover from the AP's language to the original language of ideologically congruent

[18] One limitation of this analysis is that the Gentzkow and Shapiro (2010) index is based on data from the year 2005. It is possible that outlets' ideological leanings have changed between that year and my sample period.

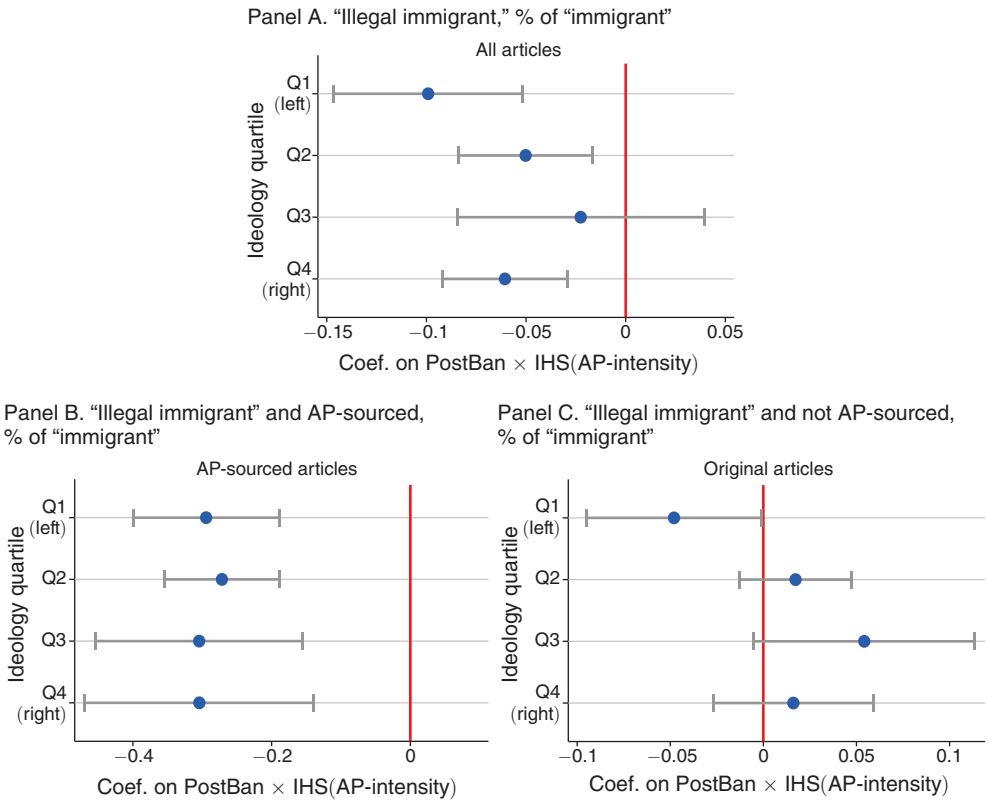

FIGURE 6. DIFFUSION OF THE BAN BY OUTLET IDEOLOGY

Notes: Effect of the AP's ban on media outlets' use of the term "illegal immigrant": Heterogeneity by Gentzkow-Shapiro index of outlet ideology. Q1 to Q4 denote quartiles of the index, with higher values indicating more right-leaning ideology. Panel A presents coefficients and 95 percent confidence intervals from regressions of the monthly use of the term "illegal immigrant" relative to the term "immigrant" by outlets in each quartile, on the interaction of IHS(AP-intensity) and PostBan. The regressions control for outlet and year-month fixed effects and are weighted by the number of articles using the term "immigrant." Panels B and C replicate panel A, splitting the dependent variable into articles that use the term "illegal immigrant" and are sourced from the AP (B) versus ones that are not (C), both relative to the number of articles that use the term "immigrant." All dependent variables are standardized to mean zero and standard deviation one within each ideology-quartile. The figures show point estimates and 95 percent confidence intervals. Standard errors are clustered by media outlet. Online Appendix Table B7 presents the corresponding regression results.

outlets, and backlash from ideologically incongruent outlets. Yet these effect are only marginally significant and small in magnitude, so that on net, the uniformly large and significant decline in the use of "illegal immigrant" in AP-sourced content dominates.[19]

[19] The raw data are also consistent with these results: Ideology appears to be correlated with the average frequency of use of the term "illegal immigrant" in original articles, but not in AP-sourced articles. Online Appendix Figure B4 suggests an ideology gradient in original content: Both before and after the ban, more right-leaning outlets use the term "illegal immigrant" more often in original articles. Yet no such gradient exists in AP-sourced content: Regardless of ideology, the frequency of use of the term is equal to that in AP dispatches ($\approx$ 50 percent before the ban and $\approx$ 5 percent after the ban).

Overall, the heterogeneity results are consistent with a degree of inertia in the use of AP copy in content production, such that the ban leads to a net reduction in use of the term "illegal immigrant," even in right-leaning outlets. This is despite their likely ideological opposition to the ban and some evidence of backlash in original content.[20]

Diffusion by Outlet Size.—On the other hand, the ban may have limited potential to affect public opinion if the diffusion effects documented above are concentrated in outlets with low readership. I test this using data on circulation in the year 2010 that are available for all print newspapers. I group newspapers into quartiles of circulation and estimate separate difference-in-differences regressions for each subsample. The results in online Appendix Table B9 suggest significant diffusion of comparable magnitude, regardless of newspapers' readership.

Other Aspects of Immigration Coverage.—In Figures B6 and B7, I examine the impact of the ban on other aspects of the coverage of immigration. I repeat the steps from the analysis of AP dispatches in Section I, but apply them to the corpus of media articles using the term "immigrant." I then run difference-in-differences regressions following specification 2, taking each of the measures of coverage as a dependent variable. Consistent with the finding that the effects of the ban are mostly driven by AP-sourced content, the results closely follow the patterns documented for AP dispatches: (i) only partial adoption of AP-approved substitutes for the term "illegal immigrant," (ii) a null effect on the volume of immigration coverage, and (iii) no impact on slant or sentiment, other than that driven by the absence of the term "illegal immigrant."

Taken together, these (null) results lend support to an interpretation of the ban as a shock to use of the term "illegal immigrant" that leaves other features of immigration coverage largely unaffected.

III. Views on Immigration Policy

Having established that the ban produces significant variation in media outlets' use of the term "illegal immigrant," in this section I exploit this variation to study how it affected public views on immigration policy. To this end, I compare pre- and post-ban responses in the CCES electoral survey in counties with different AP intensity of locally circulated newspapers.

A. *Data*

Aggregation to the County Level.—Since the CCES does not ask *which* media outlets the respondent reads, I rely on the county of residence and assess exposure to locally circulated newspapers. The first step in this analysis is to aggregate my

[20] Another margin along which the ban could have faced resistance is the response of readers, who may reduce their readership if the ban is incongruent with their political views (Durante and Knight 2012). Yet online Appendix Table B8 suggests no significant effect of the ban newspapers' circulation—either on average or for right-leaning newspapers, whose readers may be most likely to respond.

measures of newspapers' content to county level. To do so, I obtain data on the geographic distribution of daily newspapers' circulation from the Alliance of Audited Media (AAM). I use their Fall 2012 "Newspaper GEO/Circ" report, which includes circulation by newspaper and zip code from the most recent audit prior to this date, and aggregate zip code-level data to county level.[21] Since the AAM does not collect geographically disaggregated data for low-circulation newspapers, I impute these observations with data on total circulation from the *Editor and Publisher* yearbooks, assuming that small newspapers circulate mainly in the county of their headquarters. This procedure is similar to the one used by Seamans and Zhu (2014).[22]

I match these data to the data on content from NewsLibrary and ProQuest based on the name, town, and state of the newspaper. I then keep counties for which newspapers matched to the NewsLibrary-ProQuest sample account for at least 90 percent of total county circulation. This ensures that county-level data on newspapers' content are measured with reasonable precision. The resulting dataset contains about 2,300 counties (out of a total of 3,000) and 800 daily newspapers (out of a total of 1,200).

I aggregate AP intensity to county level by averaging the AP intensity of newspapers circulated within the county (in number of AP-sourced articles per 1,000), weighting each newspaper by its county-specific circulation. Formally:

$$APintensity_c = \frac{\sum_m \left(circ_{mc} \times APintensity_m\right)}{\sum_m circ_{mc}}, \tag{3}$$

where $circ_{mc}$ is circulation of newspaper m in county c. Online Appendix Figure C1 presents the resulting geographic distribution of AP intensity. Similarly, I aggregate the circulation-weighted share of articles that use the term "illegal immigrant," relative to ones that use the term "immigrant," by county and year:

$$Illimm/Imm_{cy} = \frac{\sum_m \left(circ_{mcy} \times Illimm/Imm_{my}\right)}{\sum_m circ_{mcy}}. \tag{4}$$

Correlates of AP Intensity.—To assess the correlates of AP intensity and control for them, I collect data on county-level demographic, economic, and political characteristics. Data on annual county population are from ICHS. Data on population density and urban share of the population are from the 2010 census. Turnout and the Republican vote share in the 2012 presidential election are from Dave Leip's Atlas (Leip 2010–2016). Racial composition, share college educated, share foreign-born and log income per capita are from the 2012 five-year American Community Survey (2012) and the 2012 unemployment rate is from the Bureau of Labor Statistics (2012). Air distance from the county centroid to the Mexican border is computed in GIS. Finally, county-level newspaper circulation per capita is estimated with data from the Alliance of Audited Media 2012 "GEO/Circ" report combined with the *Editor and Publisher* 2010 yearbook.

[21] For the largest nationally circulated newspapers the AAM only reports circulation at DMA level. For these cases I assign circulation to counties in proportion to adult population (i.e., over the age of 18).

[22] Similar results obtain using data on circulation by newspaper and county compiled by Snyder and Strömberg (2010). This dataset is based on circulation reported to the Audit Bureau of Circulation (ABC, which later became "Alliance for Audited Media"), combined with the Standard Rate and Date Service for non-ABC newspapers. The drawback of this dataset is that it only covers the period up to 2006.

Online Appendix Figure C2 presents the univariate correlations of each of these variables with AP intensity. AP intensity is significantly positively correlated with Republican vote share and significantly negatively correlated with the share of college educated, the share of foreign-born, and county-level newspaper circulation. This is consistent with the notion that smaller newspapers are more likely to resort to sourcing content from the AP rather than producing original reporting. A higher-educated audience, as well as one with more immigrants, may also have higher demand for original content, particularly on immigration.[23]

The CCES Survey.—To assess how public opinion on immigration policy changed in response to the ban, I use a large nationally representative survey: the *Cooperative Congressional Election Study* (CCES) (Kuriwaki 2022). The CCES is a repeated cross-section survey with 50 to 60,000 respondents per wave in election years and smaller waves of 10,000 to 20,000 respondents in nonelection years. The survey is administered online, and a large portion of participants are YouGov panelists. Importantly for my setting, a large share of survey respondents (33 percent) report that they regularly read a newspaper in print i.e., that they did so on the day before they were surveyed.

Views on Immigration Policy.—CCES respondents are asked the multiple choice question "What do you think the US government should do about immigration?" The set of possible answers differs by survey wave. Online Appendix C.3 presents all options and lists the waves they appear in, following the harmonization provided by Dagonel (2021). Five policies are included in at least one survey wave before the ban, and at least one survey wave after the ban:

1. increase the number of border patrols on the US-Mexican border;
2. grant legal status to all illegal immigrants who have held jobs and paid taxes for at least three years, and not been convicted of any felony crimes;
3. fine US businesses that hire illegal immigrants;
4. allow police to question anyone they think may be in the country illegally;
5. prohibit illegal immigrants from using emergency hospital care and public schools.

I examine each of the five questions separately and construct a dummy variable equal to one if the policy was selected. Since the first option is the only one that does not include the term "illegal immigrant" in its text, and is asked consistently in all CCES waves, I consider it as the main outcome. Alternatively, I aggregate the two options included consistently in all survey waves (items 1 and 2) into an index. I also aggregate all thirteen immigration-related questions, including ones included in a single survey wave, into a second index. The indexes are constructed by recoding the response to each question in the direction of restricting immigration

[23] Online Appendix Figure C3 compares the correlates of Reuters intensity—which is based on credit in all articles—to the correlates of the equivalent measure of AP intensity. The figure suggests that Reuters-intensive and AP-intensive counties share some similar characteristics. For example, both variables are strongly negatively correlated with newspaper circulation per capita, and for both variables this is the main correlate in terms of magnitude.

and taking the average across standardized responses (following Kling, Liebman, and Katz 2007).

Views on other Policies and Voting.—As placebo outcomes, I also use CCES questions that relate to policy issues other than immigration. I consider all policy questions that appear on the questionnaire at least once before the ban and at least once after. Specifically, I create dummies for support for always allowing women access to abortion, opposition to gay marriage, support for repealing the Affordable Care Act, opposition to affirmative action, preference for spending cuts over tax increases, preference for sales taxes over income taxes, support for foreign military intervention, preference for jobs over the environment, and the opinion that the economy has gotten worse over the past year (see online Appendix C.3 for details). I aggregate these variables into a placebo index, using the same method as for views on immigration policy.

The survey also asks about the respondent's vote (if surveyed after an election) or voting intention (if surveyed before an election) in presidential, Senate, and House elections, as well about the respondent's approval of the incumbent president. I code dummy variables equal to one if the respondent voted or intends to vote for a Republican candidate and record disapproval of Obama/approval of Trump on a four-point scale.

Finally I obtain county-level Republican vote shares for the 2012 and 2016 presidential elections i.e., the shares of votes for Romney and Trump, respectively, as well as the Republican vote share in House and Senate races from David Leip's Election Atlas.

B. *Empirical Strategy*

To identify the intention-to-treat effect of the ban on views on immigration policy, I implement reduced-form difference-in-difference regressions with continuous treatment of the following form:

$$X_{icy} = \alpha_c + \beta_y + \rho IHS(APintensity_c) \times PostBan_y + \epsilon_{icy}, \tag{5}$$

where X_{cy} denotes the immigration policy preferences of respondent i in county c and year y, $APintensity_c$ is the AP intensity of newspapers circulated in county c, $PostBan_y$ is an indicator equal to one for survey waves carried out after the ban, and α_c and β_y are county- and survey-year fixed effects, respectively. Standard errors are clustered by county.

To identify the local average treatment effect of exposure to the term "illegal immigrant" in local media, I estimate 2SLS equations of the form:

$$X_{icy} = \mu_c + \nu_y + \gamma \widehat{Illimm/Imm_{cy}} + \eta_{icy}, \tag{6}$$

$$Illimm/Imm_{cy} = \alpha_c + \rho_y + \gamma IHS(APintensity_c) \times PostBan_y + \epsilon_{cy}. \tag{7}$$

The first-stage equation has the same form as in Section II, but is now estimated at the county times survey-year level (instead of at the level of media outlet times month). The excluded instrument for the potentially endogenous use of the term "illegal immigrant" relative to the term "immigrant" by local media—$Illimm/Imm_{cy}$—is the interaction of AP intensity with an indicator for the period after the ban. This approach is thus akin to a shift-share strategy in which *PostBan* is an aggregate shock and $IHS(APintensity_c)$ is local exposure to that shock.

I estimate the two sets of equations at respondent level, controlling for respondent characteristics that are likely to correlate with views on immigration policy: age, age squared, gender, and categories for education, race, immigration status, and household income.[24]

The identifying assumption is that policy views in counties with high versus low AP intensity would have evolved similarly in the absence of the ban. Although these counties are likely different (as suggested by online Appendix Figure C2), any time-invariant county characteristics correlated with AP intensity are absorbed by county fixed effects. Therefore, if unobserved county characteristics are to confound my results, their effect on policy views would have to *change* at the same time as the ban took effect. To account for this possibility, I examine the sensitivity of the estimates to controlling for a host of observed county characteristics (detailed in Section IIIA) measured at baseline and interacted with year fixed effects. I also consider specifications that restrict the identifying variation within a state or within a media market by including state $\times$ year or DMA $\times$ year fixed effects. This absorbs any unobserved confounders that vary by state or media market. My most demanding specification controls for county-specific linear trends. This allows counties with different AP intensity to be on different trends and identifies the change in views off of the sharp change at the time of the ban.

Additionally, a strict causal interpretation of the 2SLS estimates requires the assumption that the ban did not affect immigration policy views through channels other than use of the term "illegal immigrant" in locally circulated newspapers. The are two possible violations of this assumption.

First, the AP intensity of other locally consumed media may be correlated with that of newspapers, or if there may be spillovers from print to other local news coverage (Widmer, Galletta, and Ash 2022). Specifications that control for DMA $\times$ year fixed effects address this concern to some extent, since media market regions are designed to capture homogeneous TV offerings and online news consumption patterns.

Second, the ban may affect other aspects of immigration coverage besides use of the term "illegal immigrant." This concern is addressed by the null results on the volume, slant, and sentiment of immigration coverage documented in Section II—yet I cannot rule out other, potentially nonmeasurable, changes in content or tone. Since the ban coincides with a decline in outlets' use of the AP (online Appendix Figure B1), the changing share of wire versus original stories may be one source of such subtle changes.

[24] The question on immigration status is not asked in the 2015 survey wave. Excluding this control and including 2015 does not affect the results, since this is a small off-election wave and accounts for only 4.5 percent of the baseline sample.

Table 4—Support for Increasing Border Security

	Increase border security				
	(1)	(2)	(3)	(4)	(5)
Reduced form					
PostBan × IHS(AP-intensity)	−0.0051	−0.0047	−0.0060	−0.0054	−0.0057
	(0.002)	(0.002)	(0.002)	(0.002)	(0.003)
County fixed effects × linear time trend	No	No	No	No	Yes
DMA × year fixed effects	No	No	No	Yes	No
State × year fixed effects	No	No	Yes	No	No
County controls × year fixed effects	No	Yes	Yes	Yes	Yes
Observations	192,635	192,072	192,072	192,068	192,072
Number of counties	2,157	2,143	2,143	2,143	2,143
R^2	0.09	0.09	0.09	0.10	0.10
Mean dep. var.	0.55	0.55	0.55	0.55	0.55
Effect for SD Δ in treatment	−0.0083	−0.0077	−0.009	−0.0087	−0.0092
2SLS					
"Illigal immigrant," percent of "Immigrant"	0.0057	0.0045	0.0071	0.0048	0.0055
	(0.002)	(0.002)	(0.002)	(0.002)	(0.003)
First-stage F stat.	10.53	27.61	22.32	29.82	11.71

Notes: Effect of the AP's ban on support for increasing border security. Reduced-form OLS regressions in the top panel and 2SLS regressions in the bottom panel. All regressions are at respondent level and control for respondent characteristics, and county and year fixed effects. Standard errors clustered by county.

In both cases, the reduced-form intention-to-treat estimates remain valid with somewhat different interpretation: They capture the combined effect of all changes in local immigration coverage induced by the ban.

Finally, equation (6) identifies a local average treatment effect: It applies to readers of newspapers that change their language on immigration solely due to the change in the input supplied by AP. Such newspapers are likely to have a less pronounced stance on immigration policy and potentially more persuadable readers, compared with the readers of always- or never-takers.

C. *Results*

First Stage.—I start by replicating the analysis of the diffusion of the ban for the new sample (print newspapers) and unit of observation (county × year). Online Appendix Figure C4 presents the first stage of equation (6). It suggest that at county level, 1-standard-deviation higher AP intensity (= 1.6 IHS-points) is associated with a reduction in use of the term "illegal immigrant" after the ban of about 8% relative to the mean.

Intention-to-Treat Effect.—I then turn to the reduced-form intention-to-treat effect of the ban on support for restrictive immigration policies (equation (5)). Table 4 presents the results for support for increasing border security. I find that respondents in counties with 1-standard-deviation higher AP intensity show about 0.8 p.p. lower support for increasing border security after the ban. This effect is robust to various alternative specifications that gradually introduce more demanding controls, including baseline county characteristics × year fixed effects, state *times* year or DMA × year fixed effects, and finally—county fixed effects × linear time

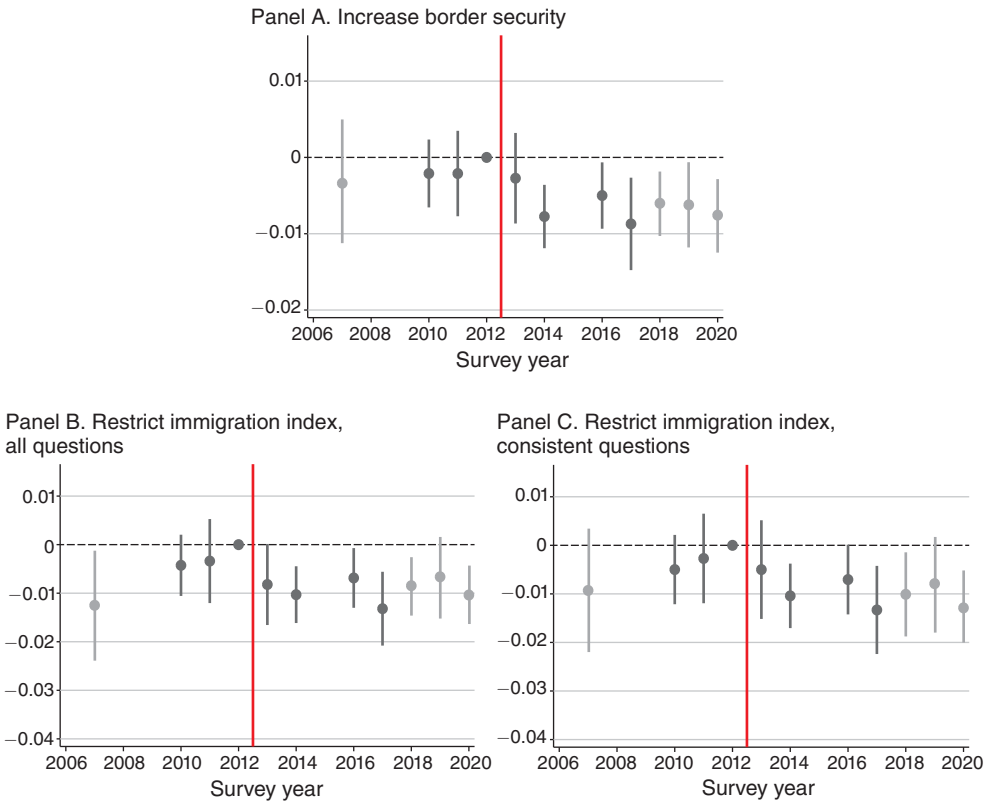

FIGURE 7. SUPPORT FOR RESTRICTIVE IMMIGRATION POLICIES: DYNAMIC DIFFERENCE-IN-DIFFERENCES

Notes: Effect of the AP's ban on support for restrictive immigration policies: Dynamic difference-in-differences. Each panel presents coefficients and 95 percent confidence intervals from a reduced-form OLS regression of a given measure of support for restrictive immigration policy, on survey year fixed effects interacted with IHS(AP-intensity). The year 2012 is the omitted category. Coefficients corresponding to survey years outside the main sample period (2009–2017) are shown in light gray. The dependent variables are: support for increasing border security in panel A, an index of all immigration-policy questions in panel B, an index of the two consistent immigration-policy in panel C. Regressions are at respondent level and control for respondent characteristics, baseline county characteristics interacted with time, and county and year fixed effects. Standard errors clustered by county. Online Appendix Table C11 presents the corresponding regression coefficients for each immigration policy question.

trend. These specifications absorb any confounders that vary by state or DMA, or are county-specific but vary smoothly over time. Online Appendix Tables C2 and C3 show that similar results obtain for the two aggregate indexes of support for restrictive immigration policy, and that the estimates are robust to the alternative specifications described above.

To examine the dynamics of the reduced-form effect, I estimate a regression that interacts IHS(AP-intensity) with indicators for survey wave, with 2012 as the baseline category. In this analysis, I can furthermore add the survey waves from 2007, and 2018–2020 to my baseline sample in order to examine longer-term trends. This comes with the caveat that the sample size in the earliest survey wave, 2007, is lower, with only 6,000 respondents. Figure 7 presents the results for support for border security and for the two aggregate indexes. The figure shows that the pretrend

Table 5—Support for Increasing Border Security: Robustness and Reuters Placebo

	Reduced form				2SLS		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Increase border security						
PostBan × IHS(AP-int.)	−0.0043	−0.0036	−0.0043				
	(0.0014)	(0.0015)	(0.0014)				
PostBan × IHS(Reuters-int.)				−0.0019			
				(0.0038)			
"Illegal immigrant," percent of "Immigrant"					0.0049	0.0040	0.0049
					(0.0018)	(0.0019)	(0.0021)
Method	Credit	Plagiarism	Credit	Credit	Credit	Plagiarism	Credit
Sample	Immigr.	Immigr.	All	All	Immigr.	Immigr.	All
Observations	192,072	192,072	192,072	192,072	192,072	192,072	192,072
Number of counties	2,143	2,143	2,143	2,143	2,143	2,143	2,143
Mean dep. var.	0.55	0.55	0.55	0.55	0.55	0.55	0.55
First-Stage *F*-stat.					20.94	21.37	15.43

Notes: Effect of AP's ban on support for increasing border security: Robustness to alternative measures of AP intensity and Reuters placebo. *Method* and *Sample* describe the construction of AP intensity (Reuters intensity) in the respective column. Reduced-form OLS regressions in the left panel and 2SLS regressions in the right panel. All regressions are at respondent level control for respondent characteristics, baseline county characteristics interacted with time, and county and year fixed effects. Standard errors clustered by county.

is, if anything, increasing in the long run, and suggests that the effects persist until 2020.

In online Appendix Figure C5, I estimate a flexible version of the reduced-form equation, splitting the distribution of AP intensity into quantiles and interacting each quintile with an indicator for the period after the ban. I find that the reduced-form effect is strongest for counties in the top quantile of AP intensity: Compared to counties in the lowest quantile, respondents in this group reduce their support for border security by 2.5 p.p.

Local Average Treatment Effect (*LATE*).—In the bottom panel of Table 4, I estimate the LATE effect of exposure to locally circulated articles using the term "illegal immigrant" (equation (6)). The results are consistent with the intention-to-treat estimates and suggest that a 10 p.p. increase in use of the term "illegal immigrant" in local media (i.e., about a 50 percent increase relative to the mean of 21 percent) is associated with an increase in support for increasing border security of about 4.5 p.p. Online Appendix Table C4 suggests comparable magnitudes for the LATE expressed in terms of different definitions or transformations of the number of articles using the term "illegal immigrant."

Robustness and Reuters Placebo.—In Table 5 and online Appendix Tables C5 and C6, I repeat the robustness checks from Section II that vary by the method used to construct AP intensity, vary the period of measurement, or use alternative transformations of the share of AP-sourced articles. All of these checks yield similar results, in both the reduced form and the 2SLS estimates.

Instead, I find no significant effect of a placebo treatment defined as the interaction of Reuters intensity and Post-Ban (column 4 of Table 5). This is reassuring,

since it suggests that the effect is specific to exposure to the AP rather than to news wires in general.

I also probe the robustness of the results to alternative ways of matching CCES respondents to newspapers. Many US newspapers nowadays are local monopolies: 40 percent of counties in my sample have only one newspaper with market share $\geq$ 10 percent. Therefore, another reasonable aggregation of AP intensity and of exposure to the term "illegal immigrant" is to simply assign to each county the single newspaper with the highest local circulation. Columns 1 and 4 of online Appendix Table C7 show that the findings are robust to this method. In the remaining columns I use the circulation-weighted aggregation, but split the sample into counties with one major newspaper versus those with several major newspapers (using the criterion of $\geq$ 10 percent market share). The effects are more precisely estimated in the former subsample.

In online Appendix Table C8, I instead vary the threshold for inclusion of a county in the sample. In the baseline sample, I require that the NewsLibrary/ProQuest data on content cover $\geq$ 90 percent of total county circulation. Lowering this threshold increases measurement error, as it leads to using data from counties for which the estimated content measures omit newspapers with larger market shares. Yet my estimates remain statistically significant as I lower the threshold, including in the case of keeping all counties.

Another concern may be that AP intensity depends on the degree of competition newspapers face (Angelucci, Cagé, and Sinkinson 2021), especially from online media and advertising platforms whose expansion coincides with the sample period. Online Appendix Table C9 addresses this concern by including additional controls for county-level internet and Craigslist penetration.[25] The results are robust to these controls.

Finally, the estimates are robust to clustering standard errors at a higher level (i.e., by DMA or state) to allow for correlation in AP intensity at that level (online Appendix Table C10).

Other Immigration Policy Questions.—Similar results obtain for other questions on immigration policy. In Table 6, I examine separately the effects on each one of the five immigration-related questions asked on the CCES at least once before and at least once after the ban. I also consider an index of support for restricting immigration that aggregates responses on all thirteen questions asked in the sample period, including rotating ones, and an index that aggregates the two consistent questions. Online Appendix Table C11 presents the corresponding reduced-form coefficients estimated separately by survey year.

With the exception of the question on legalization, the results suggest a significant effect of the ban on support for restrictive immigration policies, in both the reduced form and 2SLS estimates. The magnitudes range from 0.6 p.p. to 0.9 p.p. reduction

[25] Internet penetration is measured as the share of households with fixed Internet access service at speed exceeding 200 kbps, using data from the FCC. This data is reported discretized into 5 categories. Data on the date of Craigslist entry into each county is from Djourelova, Durante, and Martin (2021). I convert it into years Craigslist has been present in a given county, coded as 0 if Craigslist is not present.

Table 6—Support for Restrictive Immigration Policies and Falsification Test

	Increase border security (1)	Grant legal status ×-1 (2)	Sanction employers (3)	Allow police questioning (4)	Prohibit services (5)	Restrict imm. index: all questions (6)	Restrict imm.index: consistent questions (7)	Index non-immigration questions (8)
PostBan	−0.0047	−0.0011	−0.0062	−0.0047	−0.0038	−0.0064	−0.0059	−0.0002
× IHS(AP-intensity)	(0.002)	(0.001)	(0.002)	(0.001)	(0.002)	(0.002)	(0.002)	(0.001)
Respondent controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	192,072	192,072	108,655	149,016	85,850	192,072	192,072	201,378
Number of counties	2,143	2,143	2,008	2,101	1,973	2,143	2,143	2,151
R^2	0.09	0.09	0.13	0.14	0.16	0.15	0.11	0.14
Mean dep. var.	0.55	−0.48	0.61	0.40	0.42	−0.00	0.00	−0.01
2SLS								
"Illegal immigrant,"	0.0045	0.0011	0.0051	0.0046	0.0033	0.0060	0.0056	0.0002
percent of "Immigrant"	(0.002)	(0.001)	(0.002)	(0.002)	(0.002)	(0.002)	(0.003)	(0.001)
First-Stage *F*-stat.	27.61	27.61	27.57	24.42	21.99	27.61	27.61	31.19

Notes: Effect of the AP's ban on policy views: Estimates by survey question / by index of questions. In columns 1 to 5, the dependent variables are based on support for individual immigration-related policies. In column 6, the dependent variable is an index aggregating views on all immigration-related policies. In column 7, the dependent variable is an index aggregating views on the two immigration-related policies featured consistently in the CCES. In column 8, the dependent variable is an index aggregating views on other, non immigration-related policies. All dependent variables are oriented such that higher values indicate more conservative views. Reduced-form OLS regressions in the top panel and 2SLS regressions in the bottom panel. All regressions are at respondent revel and control for respondent characteristics, baseline county characteristics interacted with time, and county and year fixed effects. Standard errors clustered by county.

in support for a given policy after the ban in counties with 1-standard-deviation higher AP intensity.[26]

Views on Other Policies.—If these results reflect a general change in political leanings that by chance happens to be correlated with AP intensity, we would expect that support for other policies endorsed by the Republican Party also shifts in the same direction. In the last column of Table 6, I present the results of a falsification exercise in which I replicate my baseline specification but replace the dependent variable by an index of nine policy items featured on the CCES at least once before and at least once after the ban. Online Appendix Table C13 shows the results for each component of the index. The questions range from views on abortion and gay marriage to support for repealing the Affordable Care Act. I find point estimates close to zero (with the possible exception of abortion) and no significant effect of the ban for either the index of placebo outcomes or for any of its components.

Was the change in immigration policy views enough to shift voting choices? The answer appears to be no. In online Appendix Table C14, I show that the ban had

[26] Online Appendix Table C12 presents the corresponding OLS results for the relationship between share of locally circulated articles using the term "illegal immigrant" and immigration policy views. With the exception of the question on access to services, the unconditional correlations are positive and highly significant throughout (Table C12, panel A): in the cross-section, local exposure to the term "illegal immigrant" is strongly associated with preferences for restrictive immigration policy. Instead, and in contrast to the LATE estimates discussed above, in specifications that are saturated with fixed effects and county controls interacted with time, I find small and insignificant OLS coefficients (Table C12, panel B). One explanation for this difference is the nature of compliance with the ban. The group of compliers consists of newspapers that only change their language due to the AP's input. In contrast to always-takers (who would have changed their language regardless) and never-takers (who do not change their language despite the ban), such newspapers are likely to have a less pronounced ideological stance on immigration in either direction, and hence, more potentially persuadable readers compared with the general public.

Table 7— Support for Increasing Border Security: Heterogeneity

	Newspaper readership			Ideology			Share foreign-born	
	Did not read newsp. (1)	Read newsp. (2)	Read print (3)	Liberal (4)	Moderate (5)	Conservative (6)	≥ p75 (7)	< p25 (8)
PostBan × IHS(AP-intensity)	−0.0009	−0.0051	−0.0068	−0.0029	−0.0071	−0.0014	−0.0001	−0.0087
	(0.003)	(0.002)	(0.003)	(0.003)	(0.003)	(0.002)	(0.004)	(0.003)
Observations	65,630	76,624	46,622	50,169	58,992	67,685	46,020	47,650
Number of counties	1,890	1,867	1,675	1,506	1,736	1,933	105	1,449
R^2	0.11	0.09	0.09	0.08	0.08	0.11	0.08	0.10
Mean dep. var.	0.54	0.56	0.59	0.31	0.52	0.77	0.49	0.59
Test for equality of coefficients	(1) vs (3): p-val = 0.127			(5) vs (4 or 6): p-val = 0.224			(7) vs (8): p-val = 0.054	
2SLS:"Illegal immigrant," percent. of "Immigrant"	0.0010	0.0056	0.0074	0.0025	0.0064	0.0014	0.0000	0.0059
	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.002)	(0.003)	(0.002)
First-Stage F stat.	19.20	16.61	16.04	22.40	27.16	26.27	9.21	36.65
Test for equality of coefficients	(1) vs (3): p-val = 0.148			(5) vs (4 or 6): p-val = 0.271			(7) vs (8): p-val = 0.099	

Notes: Effect of the AP's ban on support for increasing border security: Heterogeneity by respondents newspaper readership (columns 1 to 3), by respondent ideology (columns 4 to 6), and by share of foreign-born population in the respondent's county (columns 7 and 8). Reduced-form OLS regressions in the top panel and 2SLS regressions in the bottom panel. All regressions are at respondent level and control for respondent characteristics, baseline county characteristics interacted with time, and county and year fixed effects.

no effect on intentions to vote for the Republican candidate in elections for various offices, and a marginally significant effect on presidential approval ratings. In online Appendix Table C15, I use electoral data rather than voting intentions reported on the CCES, and confirm the null effect.

Heterogeneity.—As a first heterogeneity check, in columns 1–3 of Table 7, I examine respondents' self-reported newspaper readership, which proxies for likelihood of exposure to the treatment. I split the sample into respondents who report that they had not read a newspaper in the past 24 hours, those who had read a newspaper either in print or online, and those who had read a newspaper in print. This analysis has the caveat that the newspaper readership question was not asked in the 2012 and 2017 waves, so that sample size and power are reduced. Yet the results suggest a stronger magnitude of the effect among (self-reported) frequent newspaper readers, compared with nonreaders (p-value = 0.127).

I next examine the heterogeneity of the effects in terms of respondent ideology and local share of immigrants, which may be informative for the mechanism behind the above result. In particular, a persuasion mechanism would suggest that respondents with weaker ideological attachment, or ones less directly exposed to immigration, would have weaker priors on immigration policy and thus their views should be more sensitive to media framing. In columns 4–6, I estimate separate regressions by respondent ideology and find that the effects are most pronounced for moderates compared with liberals or conservatives (p-value = 0.224). In columns 7 and 8, I restrict the sample to respondents in counties in the top versus bottom quartile with respect to the 2012 share of foreign-born population. The results point to stronger effects in places with few immigrants (p-value = 0.054).

D. *Magnitudes*

In this section I discuss the magnitude of the estimated effect on immigration policy views. Expressed in terms of 1-standard-deviation higher AP intensity, the estimated first-stage effect at the county × year level amounts to a 8 percent reduction in the share of articles using the term "illegal immigrant." For comparison, the average circulation-weighted number of articles using the term "illegal immigrant" before the ban is 6 per month (relative to 19 articles per month using the term "immigrant"). We can hence think of this effect as a reduction of 0.5 articles per month, with the caveat that coverage in NewsLibrary and ProQuest is not universal, so that this should be taken as a lower bound.

The corresponding intention-to-treat effect on support for border security is 0.8 percentage point in the sample of all survey respondents, or 0.9 percentage point in the sample of regular newspaper readers. One way to benchmark this magnitude is to compare it to the gap between Republican and Democrat respondents in the CCES. Among Republicans, 76 percent support increasing border security, and this is the case for 37.6 percent of Democrats. Hence, the intention-to-treat effect corresponds to about 2 percent of the partisan gap.

To facilitate comparison with other studies in the media literature, it is also useful to express these magnitudes in terms of a persuasion rate. The persuasion rate is defined as the share of people who change their behavior— or, in this case, change their survey answer—in response to the treatment out of the ones who could have done so (DellaVigna and Kaplan 2007; DellaVigna and Gentzkow 2010). The persuasion rate for this treatment i.e., the share of respondents who are dissuaded from supporting restrictive immigration policy—can be expressed as

$$f = \frac{db}{de}\frac{1}{1-b_0}, \tag{8}$$

where b is support for restricting immigration, e is exposure to articles using the term "illegal immigrant," and b_0 is the share of the population that would oppose restrictive immigration policy in the absence of the treatment. With the coefficient estimated for the sample of all respondents, and taking into account that about one-third report that they read a newspaper and an average of 55 percent support increasing border security, this implies a persuasion rate of $f = (0.008)/(0.33 * 1) * (1/0.55) \approx 4.4\%$. With the coefficient estimated from the sample of newspaper readers, the implied persuasion rate is $f = (0.011)/(1 * 1) * (1/0.59) \approx 1.9\%$. Here, as standard in calculations of persuasion rates in the media literature, I am assuming that a newspaper reader reads every article. Relaxing this assumption e.g., assuming that only a fraction of articles are actually read—would lead to a higher persuasion rate. On the other hand, there is some evidence that the ban is more salient when it comes to headlines. Assuming that readers pay disproportionate attention to headlines would therefore lead to a lower persuasion rate.

This magnitude is in the lower range of effects estimated in the media literature (DellaVigna and Gentzkow 2010), consistent with the milder nature of the treatment. For comparison, Chiang and Knight (2011) estimate a persuasion rate of 2 percent for the effect of an expected newspaper endorsement and 6.5 percent for the effect of a surprising endorsement on voting intention for the respective candidate, and

DellaVigna and Kaplan (2007) estimate a persuasion rate of 12 percent for the effect of access to Fox News on voting for the Republican party.

The magnitude is also lower compared to estimates from survey experiments on the effects of framing immigrants' legality (online Appendix Table C16). For example, estimates from Rucker, Murphy, and Quintanilla (2019) suggest a persuasion rate of 43.5 percent for the (negative) effect of using the term "immigrant" instead "illegal immigrant" on prejudice, punitive behavior, and restrictive policy preferences. Estimates from Merolla, Ramakrishnan, and Haynes (2013) suggest a persuasion rate of 50.8 percent for the (negative) effect of the using the phrase "opportunity to become citizen" instead of "amnesty" on support for the policy.

Finally, it is important to note that all of the magnitudes discussed above are based on difference-in-differences regressions that, by definition, miss any spillovers of the ban on non-AP-intensive outlets and counties. In that sense, they are likely to underestimate the general impact of the ban.

E. *Mechanisms*

Persuasion versus Social Signaling.—I interpret the results discussed above as consistent with a persuasion mechanism in which exposure to the term "illegal immigrant" affects intrinsic views on immigration policy. An alternative interpretation is that the decline in exposure to the term "illegal immigrant," or the announcement of the ban itself, served as a signal that expressing anti-immigration views is no longer politically correct, and made people less likely to state such views in a survey (Braghieri2021; Bursztyn, Egorov, and Fiorin 2020). While I cannot rule it out, this mechanism is less consistent with the results for three reasons.

First, views are not expressed publicly: The CCES is administered online and anonymously (through YouGov.com). Second, it is unlikely that the ban itself served as a social signal: The data suggest that publicity about its announcement is low and uncorrelated with AP intensity. In the entire sample of 1 million articles using the term "immigrant," I find only 37 articles that cover the ban, and their correlation with AP intensity is close to zero. Third, even if the social signal works through lower exposure to the term "illegal immigrant," rather than through awareness of the ban, it is not obvious how this channel would explain the heterogeneous effects for moderates and in counties with few immigrants (Table 7). This heterogeneity is, on the other hand, very much consistent with the predictions of a persuasion mechanism.

Equivalence versus Emphasis Framing.—The natural experiment created by the AP's ban is a mix of two distinct treatments: (i) "Illegal immigrant" is replaced by "[immigrant who] entered the country illegally," and (ii) "illegal immigrant" is replaced by "immigrant," without any reference to legal status. Using the terminology of Druckman (2001), the two treatments can be classified as (i) an *equivalence frame*, i.e., the two terms are logically equivalent but differently phrased and (ii) an *emphasis frame*, i.e., the two terms differ in emphasizing (or omitting) a different subset of potentially relevant considerations. While the setting does not allow me to disentangle which treatment is driving the results, my interpretation leans toward emphasis framing as a main mechanism, for two reasons.

First, as documented in Sections I and II, the substitution intended by the AP was very imperfect in practice, so that the equivalence frame accounts for a smaller share in the mix of treatments. In AP content, use of the AP-approved substitutes compensates for about 13 percent of the decline in use of the term "illegal immigrant" (Figure 2). In terms of the impact of the ban on media outlets, the effect size is between 15 percent and 17 percent of that for the decline in use of "illegal immigrant" (comparing Tables 2 and online Appendix Table B6).

Second, survey experimental studies strong impacts of emphasis frames on views toward immigrants and immigration policy, while the evidence for equivalence frames is less conclusive. Online Appendix Table C16 summarizes the findings of five such studies. Of three interventions on legality emphasis frames (e.g., "immigrant" versus "illegal immigrant"), all three find that emphasizing legal status has strong negative effects on views or shifts policy preferences in a more conservative direction. The respective persuasion rates range from 14 percent to 51 percent.[27] Of six interventions using equivalence frames (usually "undocumented immigrant" versus "illegal immigrant"), three find significant effects in the expected direction, and three find null or opposite effects.

IV. Conclusion

This paper documents a large degree of diffusion of language from news wires to media outlets. Changes in news wire language rules, which are determined centrally rather than in response to the political leanings of the owners or readers of a particular media outlet, can therefore be a useful source of variation to estimate the effects of slanted language in the mass media on public policy views.

By applying this strategy and exploiting the particularly salient ban on the term "illegal immigrant" by the AP, I find evidence consistent with the ban leading to lower support for restrictive immigration policies. This effect is more pronounced for news consumers with weaker ideological attachments and lower direct exposure to immigration, consistent with a persuasion mechanism.

This evidence provides proof of concept for a key argument in recent discussions of politically correctness: That our perception of others can be influenced by the language we hear used about them. Yet the evidence in this paper is limited to the setting of unauthorized immigration and to exposure to one specific term. More work is needed to understand the external validity of slanted language as a mechanism of persuasion.

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[27] Relatedly, Flores and Schachter (2018) show that people systematically associate unauthorized status with lower social class and with criminal background, even when such associations are not borne out in reality. For example, immigrants with criminal records are judged as 18 p.p. more likely to be unauthorized. In reality, crime rates for unauthorized immigrants have been shown to be *lower* than those for natives and authorized immigrants (Light, He, and Robey 2020).

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